Graphing Sine and Cosine Functions

Name _____

Algebra 2 Lesson 7-3 (fast track) Date _____ Period _____

Learning goal: I can find the amplitude and period of a sine or cosine function from its equation or its graph, and use them to compare two periodic functions.

Language goal: I can explain how two periodic functions compare using the words *amplitude*, *period*, *midline*, and *frequency*.

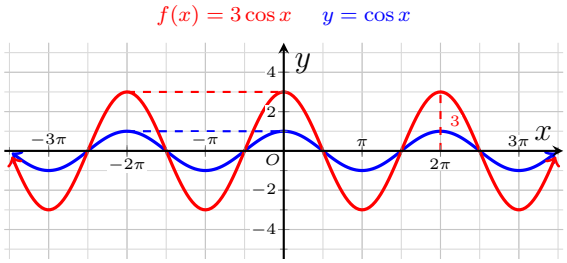
On the quiz: (1) True amplitude and period statements: Example 3. (2) Rate of change (full Savvas quiz only). (3) An equation against a graph: Example 6. (4) Reflect $f(x) = 5\cos(2x)$ over the x-axis: Example 3B, Practice. (5) Tide amplitude (full Savvas quiz only).

Example 3: Identify Amplitude and Period

For $y = a\sin bx$ or $y = a\cos bx$ $(a \neq 0, b \neq 0)$: the **amplitude** is $|a|$ (the distance from the **midline** to a maximum or minimum) and the **period** is $\frac{2\pi}{|b|}$ (the length of one cycle).

A. What are the amplitude and period of $f(x) = 3\cos x$?

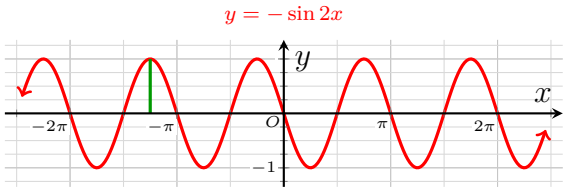
1. Graph $f(x) = 3\cos x$ and the parent $y = \cos x$.
2. Midline $y = 0$. Maximum values 3, minimum values -3: 3 times those of $\cos x$.
3. The amplitude of $f(x) = 3\cos x$ is 3, and the period is 2π.



Amplitude 3, period 2π.

B. What are the amplitude and period of $y = -\sin 2x$?

1. $y = -f(x)$ is a **reflection** of the parent $y = \sin x$ over the x-axis.
2. $y = f(2x)$ is a horizontal compression by a factor of $\frac{1}{2}$.
3. Neither one changes the amplitude: $|-1| = 1$.
4. Period compressed by $\frac{1}{2}$: $\frac{1}{2}(2\pi) = \pi$.
5. Check with a graph:



Repeats every π. **Amplitude 1, period π.**

Say it: The amplitude of $y = -\sin 2x$ is _____ because the midline is _____ and a maximum is _____ above it. The period is _____ because

Try It 3: What are the amplitude and period of each function?

a. $y = \frac{1}{3}\cos\frac{1}{2}x$

amplitude: _____ period: _____

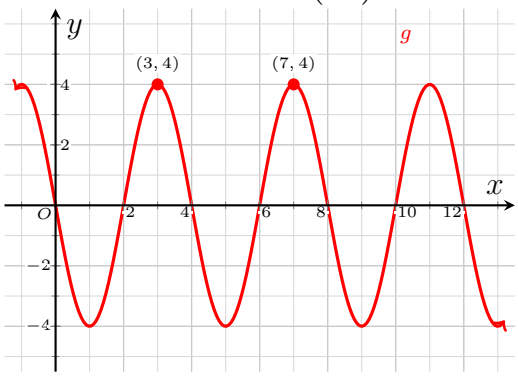
b. $y = 2\sin \pi x$

amplitude: _____ period: _____

Example 6: Compare Key Features of Two Periodic Functions

The equation for f and the graph of g are given below. How do the period and amplitude of the functions compare?

$$f(x) = 4\cos\left(\tfrac{\pi}{4}x\right)$$



f: from the equation	g: from the graph
Period: divide the parent period by the parameter $\frac{\pi}{4}$: $\frac{2\pi}{\pi/4} = 2\pi \cdot \frac{4}{\pi} = 8$	Period: measure from one maximum to the next: $7 - 3 = 4$
Amplitude: multiply the parent amplitude by the parameter 4: $1(4) = 4$	Amplitude: the midline is $y = 0$, so the distance to a maximum is $4 - 0 = 4$

Compare: the two functions have the **same amplitude**, 4, but the **period of** f (8) is **twice as long** as the period of g (4).

Study tip: use the two maximum points (or two minimum points) closest to each other to find the period.

Say it: The two functions have the same _____, but the _____ of f is _____ as long as the _____ of g, because

Try It 6: Use f and g from Example 6

a. How do the frequencies of f and g compare?

Hint: frequency is the reciprocal of the period.

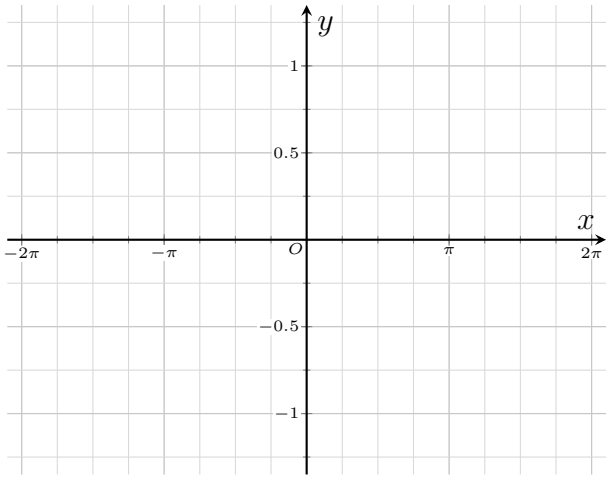
frequency of f: _____ frequency of g: _____

b. What else is different about the two functions? Explain.

Practice: DOK 3, Performance Task (Practice 23)

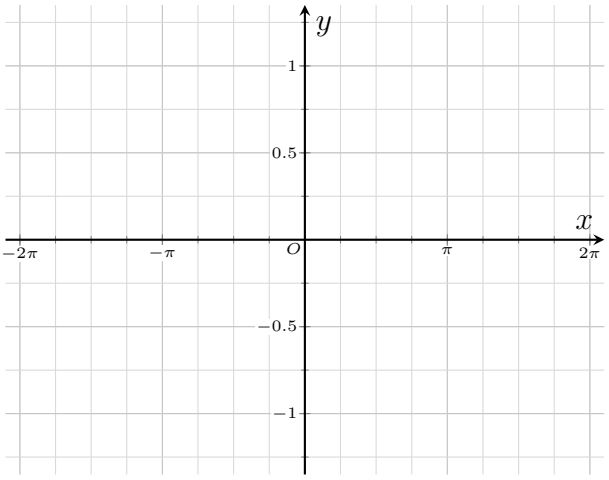
Dominga is investigating how the signs of the parameters a and b create transformations of the sine function.

Part A Graph $f(x) = \sin 2x$ and $g(x) = -\sin(2x)$ on the same coordinate plane.



Part B How are the graphs of $f(x) = \sin(2x)$ and $g(x) = -\sin(2x)$ related?

Part C Graph $f(x) = \sin(2x)$ and $h(x) = \sin(-2x)$ on the same coordinate plane.



Part D How are the graphs of $f(x) = \sin 2x$ and $h(x) = \sin(-2x)$ related?

Hint: for Parts A and C, one period is $\frac{2\pi}{2} = \pi$. Divide it into fourths and plot a zero, a maximum or minimum, a zero, the other one, and a zero.

Part E How is the graph of $y = a\sin(bx)$ affected when a or b is replaced with its opposite? Explain.

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
6H8	Evaluate trigonometric ratios using reference angles: sin, cos, tan	Warm-up: the sine and cosine values the graphs are built from (Lesson 7-2)
EDC	Convert between radians and degrees	Warm-up: radian periods such as 2π, π, 4π (Lesson 7-1)
9S8 ★	Find properties of sine and cosine functions	Quiz 1 and 3: amplitude $\lvert a\rvert$ and period $2\pi/\lvert b\rvert$ from the equation (Example 3)
5XN ★	Write equations of sine and cosine functions from graphs	Quiz 3: amplitude and period read off a graph (Example 6); Quiz 4 bonus: a negative a reflects over the x-axis
A7V	Graph sine and cosine functions	Optional: Practice Parts A and C; feeds no scored quiz item

Before the quiz, finish these sentences. (Spanish: amplitude = *amplitud*, period = *período*, midline = *línea media*, frequency = *frecuencia*, periodic function = *función periódica*.)

The _____ is the distance from the midline to a maximum or a minimum, so the amplitude of $y = a\cos bx$ is _____.

The _____ is the length of one cycle, so the period of $y = a\sin bx$ is _____.

The _____ is the reciprocal of the period, so a function with period 4 has frequency _____.

Changing a to $-a$ does not change the amplitude; it _____.

Exit: Compare $f(x) = 3\cos x$ and $y = -\sin 2x$ from Example 3. The _____ of f is _____ times as great, and the _____ of f is _____ as long, because